

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024

RI2106-1-DRIVE SYSTEMS FOR ROBOT

(RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. What is Pascal's Law

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| A) Pressure applied at a point in a confined fluid is transmitted equally in all directions. | B) The volume of a given mass of gas at constant pressure is directly proportional to the absolute temperature. |
| C) The volume of a given mass of gas at constant temperature varies inversely with the pressure. | D) The pressure of a given mass of gas varies directly with the absolute temperature, when the volume is kept constant. |

2. Pascal's Law is applicable to which of the following

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| A) Gases only | B) Liquids only |
| C) Both gases and liquids | D) Neither gases nor liquids |

3. Which type of hydraulic pump uses the principle of pushing the fluid by the back and forth movement of plunger inside a cylinder

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|----------------|---------------|
| A) Gear Pump | B) Vane Pump |
| C) Piston Pump | D) Screw Pump |

4. A hydraulic pump has a displacement of $0.01 \text{ m}^3/\text{rev}$ and operates at a speed of 1500 rpm. What is the flow rate in liters per minute

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|--------------|----------------|
| A) 15 L/min | B) 1500 L/min |
| C) 150 L/min | D) 15000 L/min |

5. 1 bar is equivalent to how many MPa

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|-------------|-----------|
| A) 0.01 MPa | B) 1 MPa |
| C) 0.1 MPa | D) 10 MPa |

6. Which device works on the principle of Pascal's Law

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|-------------------|----------------|
| A) Barometer | B) Manometer |
| C) Hydraulic lift | D) Thermometer |

7. In a hydraulic system, if the cross-sectional area of the output piston is 10 times that of the input piston, then the output force is

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|---------------------------------|-------------------------------|
| A) The same as the input force. | B) 10 times the input force. |
| C) Half of the input force. | D) 100 times the input force. |

8. Which language is commonly used for PLC programming

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| A) Python | B) Ladder Logic |
| C) Java | D) C++ |

9. What does PLC stand for in the context of industrial automation

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|----------------------------------|--------------------------------|
| A) Programmable Logic Controller | B) Programmable Linear Circuit |
| C) Portable Logic Circuit | D) Portable Linear Controller |

10. What is the primary function of a hydraulic motor in a hydraulic system

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|---|--|
| A) To convert hydraulic energy into mechanical energy | B) To store hydraulic energy for later use |
| C) To convert mechanical energy into hydraulic energy | D) To dissipate heat generated by the hydraulic system |

11. Which of the following is not a type of hydraulic motor
 A) Gear Motor B) Piston Motor
 C) Vane Motor D) Induction Motor
12. A hydraulic pump has an efficiency of 62.87%. What is the output power of the pump if the input power is 100 kW
 A) 62.87 kW B) 159.06 kW
 C) 37.13 kW D) 6.29 kW
13. What is the source of power in a pneumatic system
 A) Compressed air B) Electricity
 C) Hydraulic fluid D) Thermal Power
14. The main purpose of an electro-pneumatic system is to
 A) Control the flow of compressed air using electrical signals B) Convert pneumatic energy into Mechanical energy
 C) Convert electrical energy into pneumatic energy D) Amplify the force of compressed air
15. VFD stands for
 A) Very Fine Drive B) Variable Air Volume
 C) Variable Frequency Drive D) Variable Frequency Motor
16. What is the main operating principle of a harmonic drive
 A) The elastic deformation of a wave generator and flexspline. B) The use of a gear train with multiple stages of reduction
 C) The use of a magnetic field to couple the input and output shafts D) The use of a centrifugal force to couple the input and output shafts
17. What are the two main types of VFDs
 A) Voltage source VFDs and current source VFDs B) Scalar VFDs and vector VFDs
 C) Open-loop VFDs and closed-loop VFDs D) Voltage source VFDs and closed loop VFD
18. What is the typical operating frequency range of a VFD
 A) 10 Hz - 60 Hz B) 20 Hz - 120 Hz
 C) 30 Hz - 180 Hz D) 40 Hz - 240 Hz
19. What type of load is a variable frequency drive typically used for
 A) Resistive loads B) Inductive loads
 C) Capacitive loads D) Impedance Load
20. What is the primary function of a variable frequency drive
 A) To control the speed of an electric motor B) To convert AC power to DC power
 C) To measure voltage and current in an electrical system D) To protect electrical equipment from power surges

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) State the governing law used in fluid power system. Derive an equation for multiplication of forces.	4	L*1	1	1
	b) The kinematic viscosity of a hydraulic fluid is 0.0001 m ² /s. If it is flowing in a 30-mm diameter pipe at a velocity of 6 m/s, find the head loss due to friction in units of bars for a 100-m smooth pipe. The oil has a specific gravity of 0.90.	6	L2	1	1
	c) With a neat sketch explain the architecture of PLC.	6	L1	2	1
2.	a) With a neat sketch explain the working principle of external gear pump.	8	L1	1	1
	b) A pump having a displacement of 14 cm ³ /rev is driven at 1440 rpm and operates against a maximum pressure of 150 bar. The volumetric efficiency is 0.9 and the overall efficiency is 0.80, calculate (i) The pump delivery in LPM (ii) The input power required at the pump shaft in kW. (iii) The drive torque at the pump shaft.	8	L2	1	1

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|-------|---|----|----|---|---|
| 3. a) | One open tank is installed in the plant of which liquid level is to be controlled. When level reaches the Level Low, Outlet flow is blocked and inlet flow is allowed until high level is achieved. And when Level High is detected, outlet flow is allowed and inlet flow is blocked. Explain the PLC circuit with the help of a ladder diagram. | 10 | L2 | 2 | 1 |
| b) | Define the following: | | | | |
| i. | Laminar Flow | | | | |
| ii. | Turbulent Flow | | | | |
| iii. | Reynold Number | 6 | L1 | 1 | 1 |

Unit – II

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|-------|---|----|----|---|---|
| 4. a) | With a neat sketch explain the significance of cushioning in cylinder. | 6 | L1 | 3 | 1 |
| b) | A hydraulic motor has a volumetric displacement of 125 cm ³ and a pressure rating of 150 bars. It receives a theoretical flow of oil of 0.0015 m ³ /s from a pump. Find the motor: a) Speed b) theoretical torque c) theoretical power. | 6 | L2 | 3 | 1 |
| c) | Draw the Symbols for the following: | | | | |
| i. | Solenoid Activated spring return 4/2 DCV. | | | | |
| ii. | Shuttle Valve | 4 | L1 | 4 | 1 |
| 5. a) | Explain the sequencing of two cylinders A and B using Electro pneumatic circuit for the cylinder sequence A+B+B-A- | 10 | L2 | 4 | 1 |
| b) | Differentiate between servo valves and proportional valves. | 6 | L1 | 4 | 1 |
| 6. a) | Explain with a neat sketch working of Pneumatic Filter. | 8 | L1 | 4 | 1 |
| b) | With a neat sketch explain the working of the following along with their ISO symbols: | | | | |
| i. | NO type of Push button Switch | | | | |
| ii. | Relay | 8 | L1 | 3 | 1 |

Unit – III

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| 7. a) | With a neat sketch explain the working principle of variable frequency drive. | 8 | L1 | 5 | 1 |
| b) | What are the different types of Harmonic drives. | 8 | L1 | 5 | 1 |
| 8. a) | What is the difference between harmonic drives and compact drive. | 8 | L1 | 5 | 1 |
| b) | With a neat sketch explain the working principle of servo drive. | 8 | L1 | 5 | 1 |

BT* Bloom's Taxonomy; L* Level; CO* Course Outcome; PO* Program Outcome
